Comparison of Surgical and Obstetric Outcomes between Two Types of Cervical Polypectomy during Pregnancy

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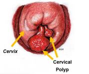
Introduction



Two main types of polyps are observed in the cervix during pregnancy: cervical (common) and decidual polyps (i.e. polypoid lesions of cervical decidua).

Cervical polyps may cause atypical symptoms such as increased vaginal bleeding, vaginal discharge, and localized infection.

Traditional polypectomy (TP) during pregnancy using forceps twisting cervical polyps, poses high risks of bleeding, preterm delivery and pregnancy loss especially for decidual polyps. **Vaginoscopy polypectomy (VP)** has emerged as an safer, less invasive, and more comfortable alternative, however its efficacy and safety compared to TP remain unclear, considering two types of polyps.

This study aims to answer (1) Do VP patients have less risk of preterm birth compared to TP patients? (2) Do VP patients have less risk of adverse events after surgery compared with TP patients? (3) Do patients with decidual polyps have a higher risk of adverse obstetric effects compared to those with cervical polyps? (4) What demographic and surgical risk factors are associated with preterm birth?

Study designs

Study design

This is a retrospective study included pregnant women who underwent cervical polypectomy at Jiangxi Maternal & Child Health Hospital between 01/2023 and 07/2024 were recruited.

Data collection

- **Baseline demographic and clinical data** were collected from patients' medical records, with consent, including current age, gestational weeks, gravidity, parity, polyp type, polyp volume, and presence or absence of chlamydia, symptom at diagnosis.
- **Surgical data** were collected from surgery records, including surgery type, surgery time, surgical bleeding.
– **Follow-up data** were collected during patients' hospital stay after underwent surgery, including total hospital stay, total hospital cost, pregnancy outcome (e.g. live birth, miscarriage, stillbirth, preterm birth, during pregnancy), gestational weeks til delivery, complications (i.e. recurrent bleeding, colpitis, high WBC)

Primary outcomes

Preterm birth: gestational weeks <37

Adverse events

Pregnancy loss: Pregnant loss unrelated to labor or induction due to fatal malformed or had to stop labor due to unclear personal issues.

Complications: Recurrent bleeding, Colpitis, High WBC (White Blood Cell Count)

Other outcomes

Surgery time, surgical bleeding, hospital stay (related to surgery), total hospital cost (related to surgery

Statistical Analysis

- We described baseline characteristics and check balance between two surgery cohort.
- We compared all outcomes (i.e. primary outcome, adverse event, other outcome) using complex descriptive table with respect to polyp and surgery type. P-values were calculated using chi-square test and wilcoxon test. Tailored bivariate table was shown.

Survival analysis

- Cohort entry was defined at gestational age 0, right censoring is recorded if participants haven't delivered baby at the end of the study (12/2024).
- We first compared risk of preterm birth using between-group bar charts and Kaplan-Meier curves
- Using the same approach, we compared risk of preterm birth with Kaplan-Meier curves between patients with deciduoid polyp and cervical polyp. We further explored whether different surgery perform different effect on participants in two polyp groups.
- We built Cox proportional hazards models and obtained unadjusted and adjusted hazard ratios with 95% confidence intervals and p-values calculated by wald test. We defined cohort entry gestational age at surgery, survival time = age at delivery - age at surgery, and right censoring as still pregnant at the end of the study or delivered full-term (>37 week)
- *** p-value<0.001, ** <0.01, * <0.05, . <0.1

Results

	Deciduoid polyp (N=75)			Cervical polyp (N=59)			P-value
	VP (N=37)	TP (N=38)	P-value	VP (N=35)	TP (N=24)	P-value	
Preterm (<37 weeks)							0.747
No	23 (62.2%)	24 (63.2%)	1	23 (65.7%)	13 (54.2%)	0.534	
Yes	14 (37.8%)	14 (36.8%)		12 (34.3%)	11 (45.8%)		
Pregnancy loss							0.249
No	27 (73.0%)	28 (73.7%)	0.534	26 (74.3%)	16 (66.7%)	0.714	
Yes	2 (5.4%)	5 (13.2%)		1 (2.9%)	2 (8.3%)		

- Baseline demographic and clinical characteristics were balances between VP and TP patients, and within each strata of deciduoid polyp and cervical polyp.
- No significant difference were found in total counts of preterm birth and pregnancy loss between two surgery type groups. No significant between-group difference in total counts of both outcomes within each strata of polyp types.

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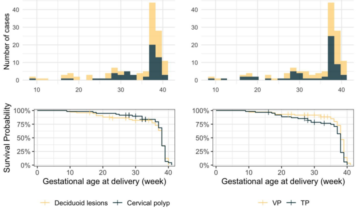
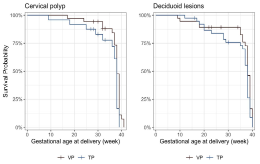
Yes	2 (5.4%)	5 (13.2%)		1 (2.9%)	2 (8.3%)		
Missing	8 (21.6%)	5 (13.2%)		8 (22.9%)	6 (25.0%)		
Recurrent bleeding(complication)							0.721
No	36 (97.3%)	37 (97.4%)	1	33 (94.3%)	24 (100%)	0.646	
Yes	1 (2.7%)	1 (2.6%)		2 (5.7%)	0 (0%)		
Colpitis(complication)							0.097
No	33 (89.2%)	38 (100%)	0.117	34 (97.1%)	24 (100%)	1	
Yes	4 (10.8%)	0 (0%)		1 (2.9%)	0 (0%)		
High WBC(complication)							0.181
No	27 (73.0%)	24 (63.2%)	0.507	26 (74.3%)	14 (58.3%)	0.315	
Yes	10 (27.0%)	14 (36.8%)		9 (25.7%)	10 (41.7%)		
Surgery time							**0.007**
Mean (SD)	16.8 (8.67)	15.8 (5.70)	0.814	13.0 (5.84)	18.6 (5.94)	**<0.001**	
Median [Min, Max]	15.0 [5.00, 47.0]	16.0 [5.00, 28.0]		12.0 [1.00, 30.0]	20.0 [5.00, 35.0]		
Surgical bleeding							**<0.001**
Mean (SD)	5.03 (4.71)	8.29 (2.91)	**<0.001**	4.63 (4.24)	9.38 (3.40)	**<0.001**	
Median [Min, Max]	5.00 [0, 20.0]	10.0 [5.00, 15.0]		2.00 [0, 20.0]	10.0 [5.00, 15.0]		
Hospital stay							**0.040**
Mean (SD)	3.73 (3.15)	4.37 (3.06)	0.225	3.43 (2.57)	4.50 (3.05)	0.077	
Median [Min, Max]	3.00 [1.00, 18.0]	4.00 [1.00, 13.0]		3.00 [1.00, 10.0]	4.00 [1.00, 14.0]		
Hospital cost							**0.013**
Mean (SD)	5990 (2050)	4730 (1870)	**0.008**	5380 (1880)	5160 (2410)	0.495	
Median [Min, Max]	5740 [2010, 12500]	4350 [1230, 8550]		5210 [2130, 10200]	4370 [1880, 12300]		

- No significant difference in total counts of adverse events were found.
- **VP patients had significantly shorter surgery time and less surgery bleeding compared to TP patients,** Among patients with deciduoid polyp, VP patients experienced less surgery bleeding (5.03 ± 4.71) than TP patients (8.29 ± 2.91) and less surgery-related hospital cost. Among patients with cervical polyp, VP patients experienced less surgery time (13.0±5.84) than TP patients (18.6 ± 5.94) and less surgery bleeding (4.63 ± 4.24) than TP patients (9.38 ± 3.40).



- From the Kaplan-Meier curves and histograms describing distribution of time to delivery between two polyp type, and between surgery type, **deciduoid group had higher risk of preterm birth (<37 week) than cervical group,** and **TP group had higher risk of preterm birth than VP group.**
- Further within each subgroup of polyp type, VP group had lower risk of preterm birth (<37 week), however the intersection between two survival curves among deciduoid subgroup suggests that VP group had higher risk of extreme preterm birth (<20 week) among those who exposed to surgery very early.



	Unadjusted HR [95% CI]	Adjusted HR [95% CI]
Age	1.001 [0.926, 1.082]	0.972 [0.876, 1.078]
Gravida (>1)	1.113 [0.631, 1.965]	0.950 [0.471, 1.919]
Para (>1)	1.638 [0.650, 4.126]	1.749 [0.542, 5.647]
Polyp volume (>2.5)	1.626 [0.890, 2.971]	1.946 [0.921, 4.116] .
Polyp type (Cervical polyp)	1.088 [0.626,1.889]	1.097 [0.562, 2.141]
Chlamydia	1.266 [0.693, 2.311]	1.317 [0.652, 2.661]
Abnormal pain(at diagnosis)	3.465 [0.835, 14.37] .	2.082 [0.260, 16.675]
Recurrent bleeding(at diagnosis)	1.429 [0.695, 2.939]	1.76 [0.714, 4.335]
Surgery type (traditional polypectomy)	1.151 [0.664, 1.993]	1.655 [0.723, 3.789]
Surgery bleeding	1.007 [0.948, 1.069]	0.955 [0.875, 1.043]
Surgery time	0.995 [0.956, 1.035]	1.021 [0.973, 1.071]
Recurrent bleeding(complication)	2.592 [0.806, 8.338]	6.135 [1.373, 27.402] *
High WBC(complication)	1.918 [1.099, 3.347] *	2.185 [1.148, 4.159] *
Colpitis(complication)	1.014 [0.246, 4.177]	1.362 [0.275, 6.741]

- **Recurrent bleeding** as a complication was strongly associated with an increased risk of preterm birth, with an adjusted HR of 6.135 (95% CI: 1.373, 27.402), indicating that patients experiencing recurrent bleeding had over six times the hazard of preterm birth compared to those without this complication.
- **High WBC** count as a complication was also significantly associated with an elevated risk of preterm birth, with an adjusted HR of 2.185 (95% CI: 1.148, 4.159). This suggests that patients with high WBC counts had more than twice the hazard of preterm birth compared to those without this complication.

Summary

Do VP patients have less risk of preterm birth compared to TP patients?

VP patients had a lower risk of preterm birth than TP patients according to the survival curves but not supported by total counts. In the decidual polyp subgroup, early surgery in VP patients was associated with a higher risk of extreme preterm birth which need to be further discussed.

Do VP patients have less risk of adverse events after surgery compared with TP patients?

No significant difference found in adverse events, but VP patients had shorter surgery times, less surgery bleeding, and lower surgery-related hospital costs than TP patients.

Do patients with decidual polyps have a higher risk of adverse obstetric effects compared to those with cervical polyps?

Decidual polyp patients had a higher risk of preterm birth than cervical polyp patients according to the survival curves, though no significant difference was observed in total counts of preterm birth or pregnancy loss between polyp types.

What demographic and surgical risk factors are associated with preterm birth?

Recurrent bleeding and high WBC count were significantly associated with an increased risk of preterm birth.

Limitation: small sample size; non-informative censoring assumption may be violated.

Conclusion

VP is an noninferior alternative to TP for pregnant women with both types of polyps.

Overall, Patients undergoing VP averagely had less surgery time, less surgery bleeding and hospital costs and potentially less risk of preterm birth. Decidual polyp patients had a higher risk of preterm birth compared to cervical polyp patients. Additionally, recurrent bleeding and high WBC counts were identified as significant risk factors for preterm birth, emphasizing the importance of monitoring and managing post-surgical complications to improve obstetric outcomes.